# Autonomous Drone with Vision-Based Object Tracking

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# Introduction

## THE CHALLENGE

Commercial leaders like DJI offer sophisticated autonomous tracking but on **proprietary, closed-source** architectures.

Open-source flight software (ArduPilot, Betaflight) exists but lacks **vision-based tracking**.

## OUR SOLUTION

- Fully autonomous drone using onboard computer vision
- YOLO11 for real-time object detection
- DeepSORT to maintain stable identity across frames
- MAVLink commands to fly and follow the target

# Detection & Tracking Pipeline

### YOLO11

Object Detection

- Single neural-network pass
- Bounding boxes + confidence
- 24 conv. layers + 2 FC layers
- Non-max suppression filtering

### DeepSORT

Object Tracking

- Kalman filter motion model
- Deep appearance descriptor
- Mahalanobis distance matching
- Stable ID across occlusions

### MAVLink

Control Protocol

- Binary serial protocol
- SET_POSITION_TARGET_LOCAL_NED
- Pixel offset → velocity cmd
- Proportional gain control (Kp)

$$e_x = x_{\text{target}} - x_{\text{center}}, \quad v_x = K_p \cdot e_x$$

### ArduPilot

Flight Controller

- Open-source autopilot firmware
- Guided Mode: external commands
- IMU + Compass + GPS fusion
- Motor throttle from velocity cmd

# Drone Architecture

## FLIGHT CONTROLLER

**ArduPilot + Guided Mode**

- Manages motors, sensors & feedback loops
- IMU (accel + gyro), Compass, GPS fusion
- Stable & level flight in Guided Mode
- Converts velocity set-points to motor throttle
- Receives commands via serial (MAVLink)

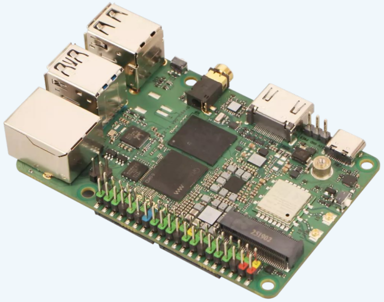
ARDUPILOT
Versatile,Trusted,Open

MAVLink

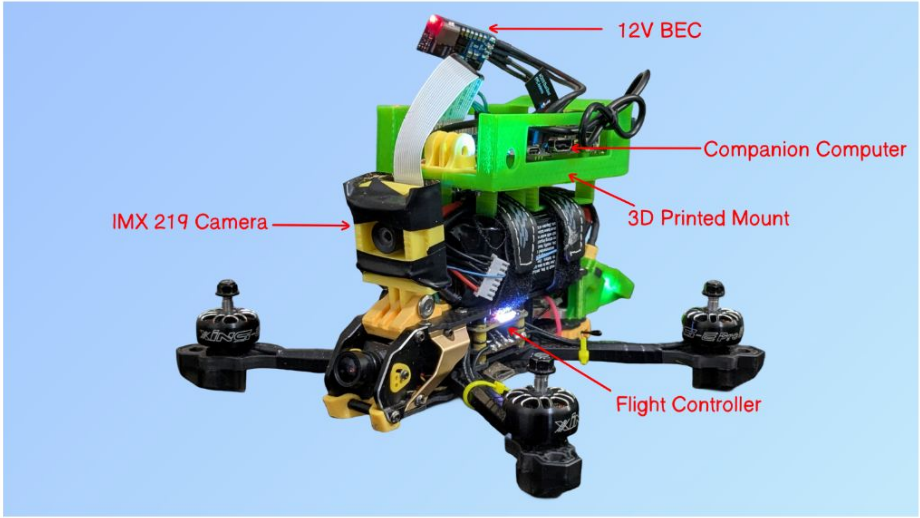
## COMPANION COMPUTER

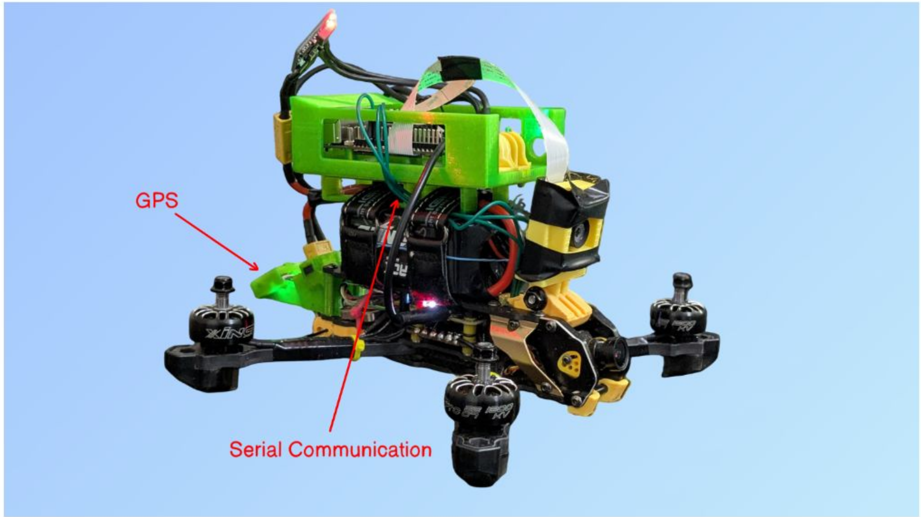
**Radxa Dragon Q6A + IMX219 Camera**

- 12 TOPS AI compute (Qualcomm QAIRT)
- Runs YOLO11 + DeepSORT pipeline
- Raspberry Pi 5 form factor (85 × 55 mm)
- Custom 3D-printed TPU mount (Fusion 360)
- GoPro-style adjustable camera enclosure

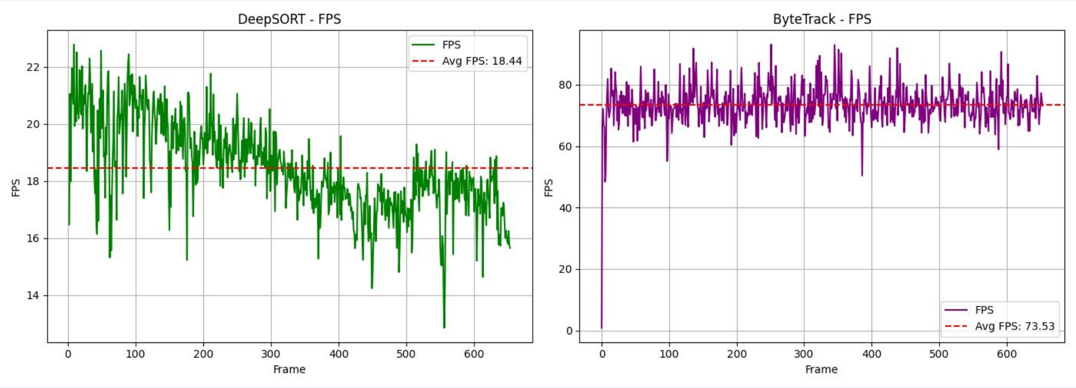


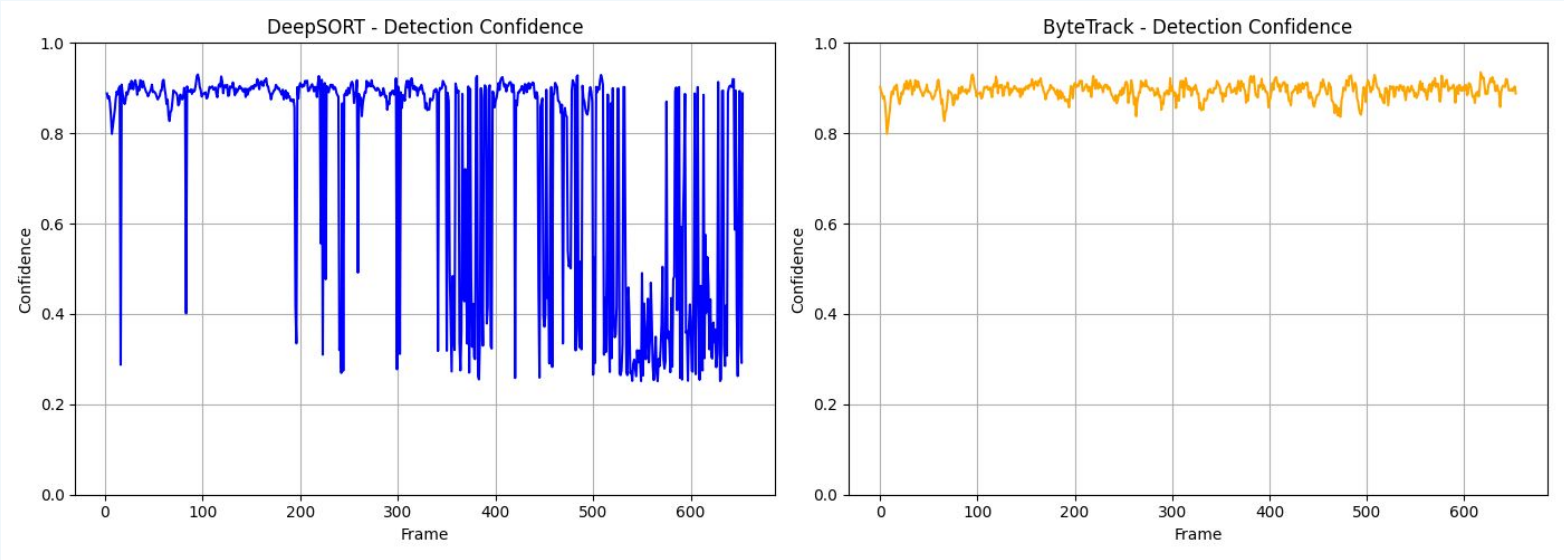
# Drone


12V BEC
Companion Computer
3D Printed Mount
IMX 219 Camera
Flight Controller

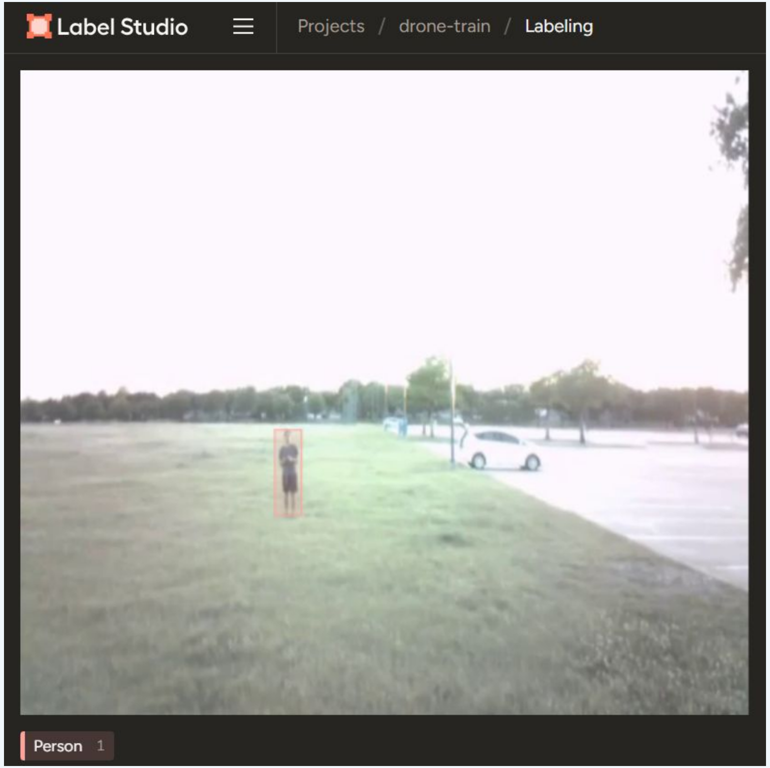

GPS
Serial Communication

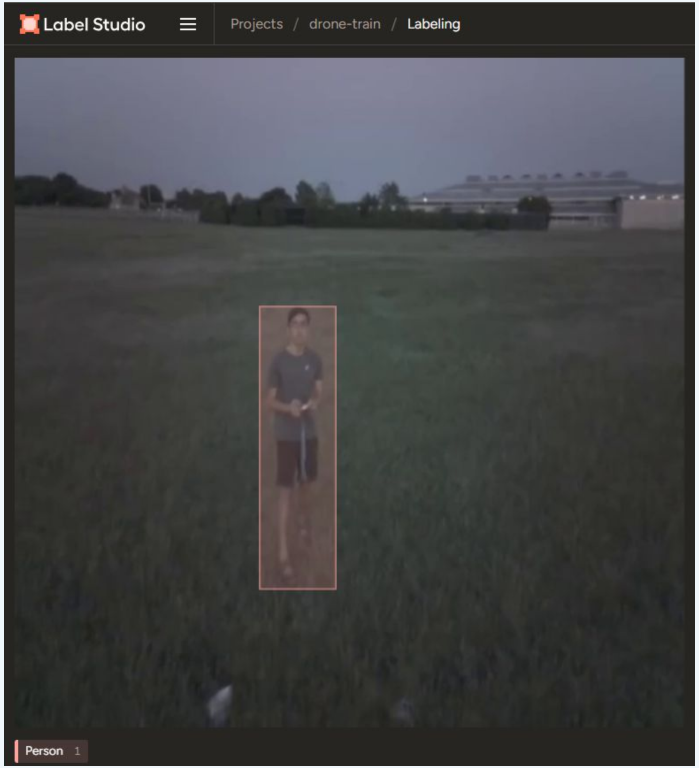
# Experiments

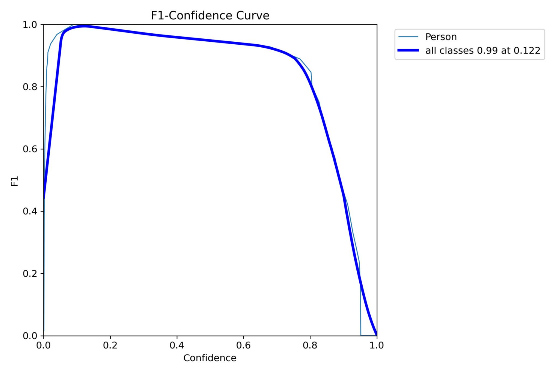
# Model Labeling and Training

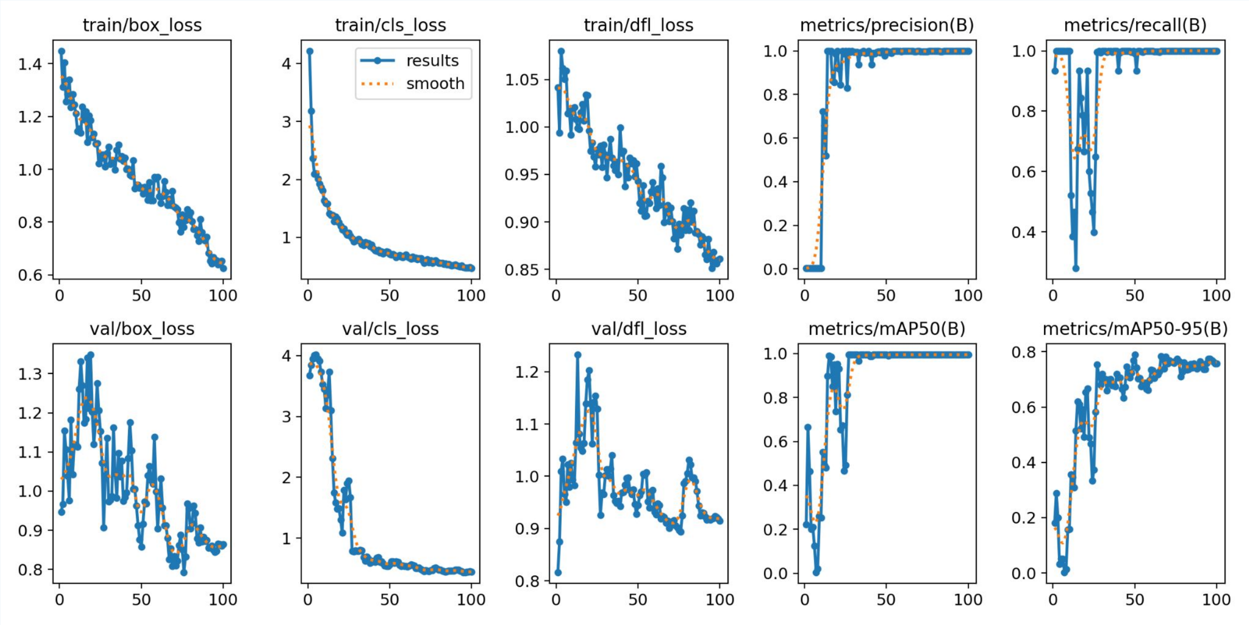



https://labelstud.io/

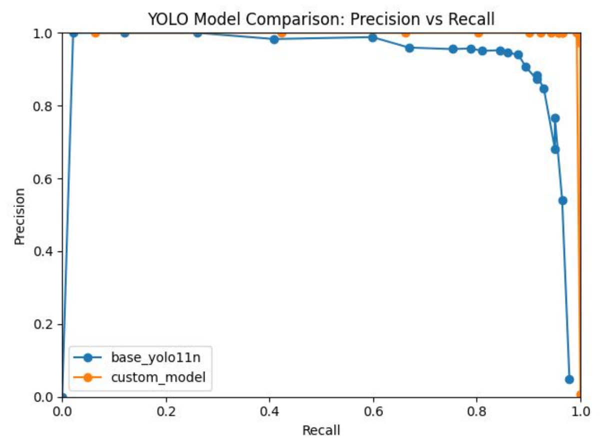
# Model Labeling and Training

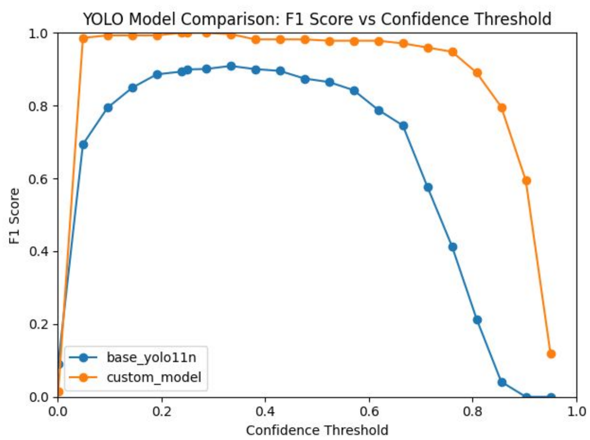


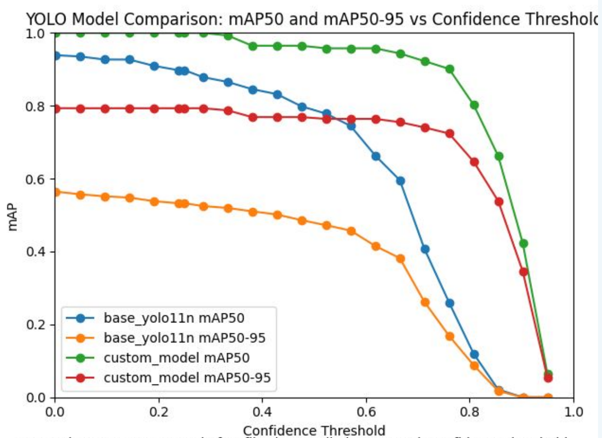




https://colab.research.google.com/github/EdjeElectronics/Train-and-Deploy-YOLO-Models/blob/main/Train_YOLO_Models.ipynb

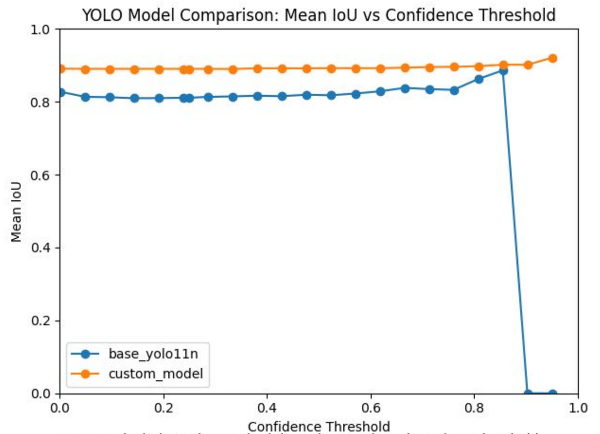
# **Results** - Yolo Base vs. Custom Trained



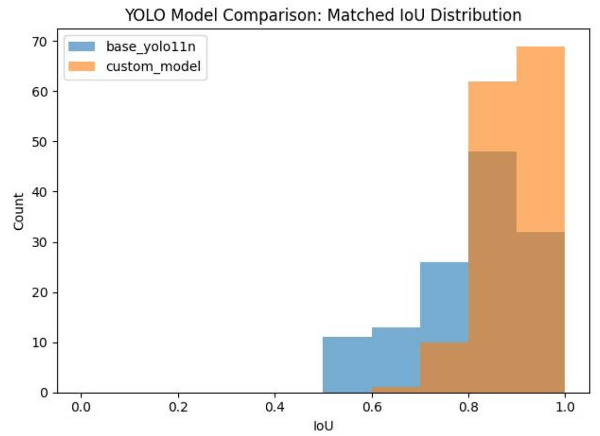




mAP values are recomputed after filtering predictions at each confidence threshold.




Mean IoU includes only matched detections at the selected IoU threshold.



# Model Optimization

PyTorch

**YOLO11n**

**NCNN**

adreno
by Qualcomm

Vulkan

# Gazebo + Ardupilot SITL Simulation

https://youtu.be/42sz89Pet-4

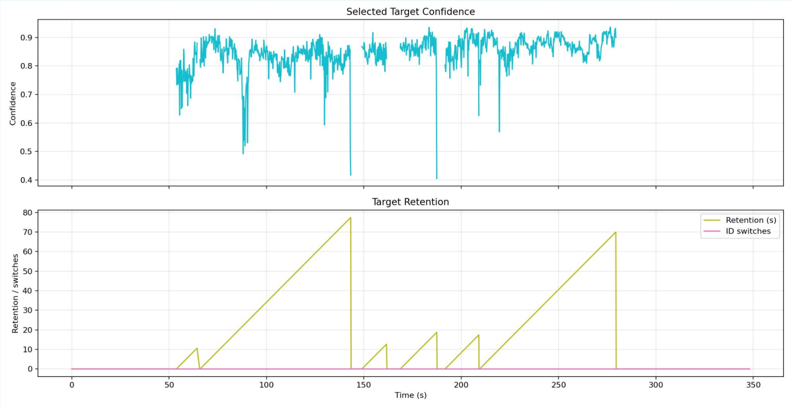
# **Real Flight** - Clip 1

https://youtube.com/shorts/hdjSHklKPtI

# **Real Flight** - Clip 2

https://youtube.com/shorts/gknYQFJ7WKk

# **Real Flight** - Results

# Questions?